Network Threats and Mitigation & Physical Security and Risk

Task 1: Create a table listing 5 common network threats (such as phishing, DDoS, man-in-the-middle, malware, and password attacks), and for each, write one real-world example from an app or service you use (like WhatsApp, Instagram, or Paytm).

No.	Threat	What it is	Real-world example
1	Phishing	Fake messages or links that trick users into giving passwords, OTPs or card details.	A fake SMS saying 'Your Paytm KYC has expired, click here' leads to a copy of the Paytm login page that steals the login and OTP.
2	DDoS	Thousands of fake requests flood a server so real users cannot get in.	When tickets for a big movie open on BookMyShow, bot traffic can flood the booking servers so the app slows down or stops working.
3	Man-in-the-Middle (MITM)	An attacker secretly sits between the user and the server and reads or changes the data.	On free public Wi-Fi at a cafe or railway station, an attacker can capture the Instagram or Paytm login data of people connected to the same network.
4	Malware	Harmful software that steals data, spies or damages the device.	A fake 'WhatsApp Plus / GB WhatsApp' APK downloaded outside the Play Store can read messages and contacts.
5	Password attack	Guessing or reusing stolen passwords (brute force, credential stuffing).	Attackers try leaked email and password pairs on Instagram logins; accounts that reuse the same password get hacked.

Task 2: Simulate a brute-force attack scenario by writing a simple Python script that tries to guess a 4-digit PIN code by checking all possible combinations against a preset value.

Constraint: The script should print the number of attempts it took to guess the correct

PIN.The script tries every PIN from 0000 to 9999 against a preset PIN (7351), counts the attempts and prints the total.

Code (brute_force.py)

brute_force.py

```
# Task 2: Brute-force simulation - guess a 4-digit PIN
import time

SECRET_PIN = "7351"   # preset value (the PIN we are trying to crack)

attempts = 0
start = time.time()

for number in range(10000):            # all combinations 0000 to 9999
    guess = str(number).zfill(4)       # keep leading zeros, e.g. 0042
    attempts += 1
    if guess == SECRET_PIN:
        print("PIN cracked:", guess)
        print("Number of attempts:", attempts)
        break

print(f"Time taken: {time.time() - start:.4f} seconds")
```

Output

Output

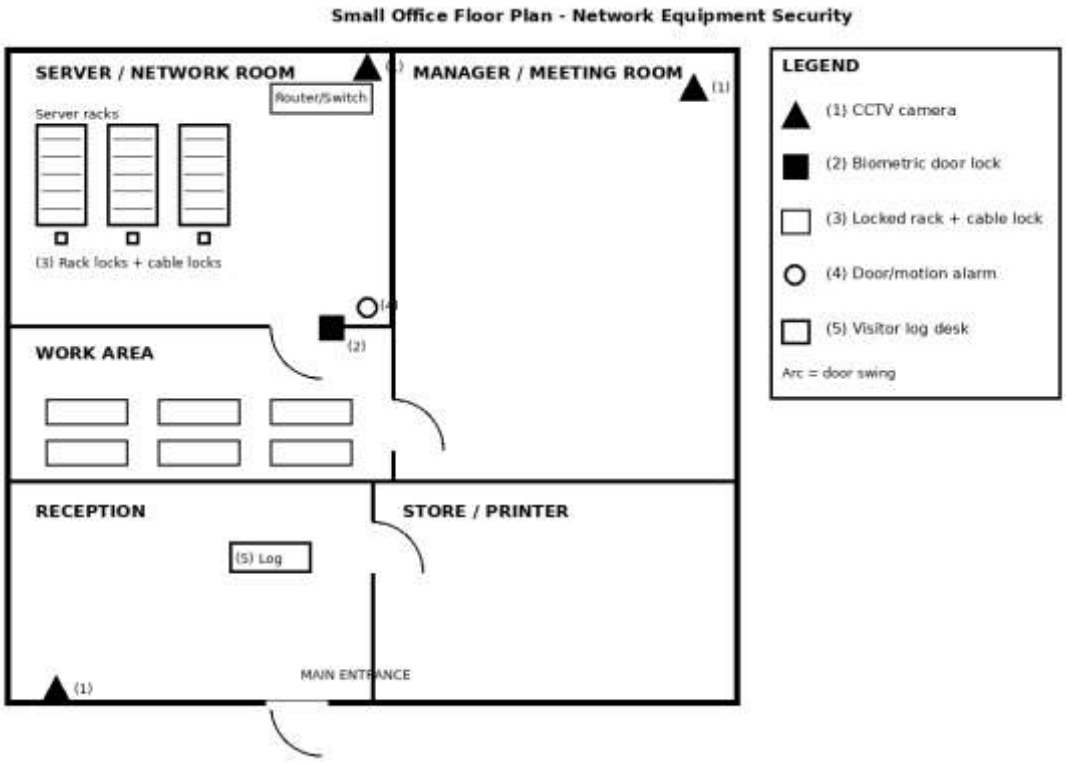
```
$ python brute_force.py
PIN cracked: 7351
Number of attempts: 7352
Time taken: 0.0016 seconds
```

Result: the PIN was cracked in 7352 attempts. A 4-digit PIN has only 10,000 combinations, so it is weak without attempt limits or lockouts.

Task 3: List 4 physical security risks that could affect a server room hosting a popular app like Zomato or BookMyShow, and suggest one mitigation step for each risk.**Example: a server room hosting an app like Zomato or BookMyShow.**

No.	Physical risk	How it can hurt the server room	One mitigation step
1	Unauthorized entry / theft	A stranger or staff member without permission steals drives or plugs in a device.	Install a biometric or card-access lock on the server room door and keep an entry log.
2	Fire	Short circuits or overheating destroy servers and data.	Fit smoke detectors and a gas-based fire suppression system (not water sprinklers).
3	Water leakage / flooding	Rain, pipe leaks or AC water damage the equipment.	Place the room above ground level and install water leak sensors with alarms.
4	Power failure / overheating	Sudden power cuts or AC failure crash the app and can corrupt data.	Use UPS plus a backup generator and temperature monitoring with alerts.

Task 4
Draw a simple floor plan (hand-drawn or digital) of a small office and mark at least 3 physical security measures you would implement to protect network equipment (such as CCTV, biometric access, or cable locks)..



Task 5: Use ChatGPT or a similar AI tool to generate a checklist of 7 actions a network admin should take to mitigate risks from both network threats and physical security breaches, then review and edit the checklist to make it specific for a college computer lab environment.

1. **Keep all lab PCs updated.** Turn on automatic OS and antivirus updates, and use a restore tool so each PC resets to a clean state after every student session.
2. **Separate the lab network.** Put lab PCs on their own VLAN, apart from staff and office systems, and let the firewall block unneeded ports.
3. **Use individual logins.** Give each student a personal account with a strong password and lock the account after 5 failed attempts. No shared logins.
4. **Lock the network cabinet and server rack.** Only the lab admin holds the key, and every opening is written in a logbook.
5. **Install CCTV.** Cover the lab entrance, the PC area and the network cabinet, and keep recordings for at least 30 days.
6. **Control ports and devices.** Disable unused switch ports and USB ports on PCs, use cable locks on equipment, and block personal laptops on the lab network.
7. **Prepare for failures.** Keep a UPS and a fire extinguisher in the lab, take weekly backups of lab data, and review network logs once a month.